

C224

Research Foundations

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A principal at a K-8 school that just opened a new cafeteria wants to understand how first-year teachers experience their mentoring. She plans in-depth interviews and will code the transcripts for themes. What kind of research is this?

- A. Experimental, since mentoring is a treatment.
- B. Quantitative, because she will count how often themes appear.
- C. Qualitative
- D. Mixed methods, since any study with more than one teacher is mixed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A researcher wants to test whether weekly homework time predicts end-of-year math scores across 600 students. Which approach fits?

- A. Quantitative correlational research
- B. Ethnography
- C. A qualitative case study of one classroom
- D. Action research in the researcher's own classroom

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A district research office with four analysts surveys 1,200 parents about homework load. Several results are surprising, so the office then interviews 15 parents to explain them. Which design is this?

- A. Explanatory sequential
- B. Exploratory sequential
- C. Convergent parallel
- D. Purely quantitative, since the survey was larger.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A third-grade teacher who coaches the school's chess club notices many students skip their reading logs. She tries a choice-board routine for six weeks, tracks log completion, reflects on the results, and adjusts the routine. What best describes her study?

- A. An experiment that proves choice boards work in every school.
- B. Survey research.
- C. Action research that improves her own practice; results are local.
- D. Not research, because teachers cannot study their own classrooms.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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5.

In a district that serves 9,000 students, two teachers score every essay with the same writing rubric. Their scores often differ by two points or more. Which kind of reliability is weak?

- A. Internal consistency
- B. Inter-rater reliability
- C. Test-retest reliability
- D. None, since rubric scores are always reliable once a rubric exists.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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6. [TRANSFER — reversed stem / different variable]

A researcher gives the same anxiety survey to the same students two weeks apart and correlates the two sets of scores. What is being checked?

- A. Inter-rater reliability
- B. Internal consistency
- C. Test-retest reliability
- D. Construct validity

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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7.

An algebra-readiness test from a vendor that also publishes textbooks is given in August. Scores correlate at $r = .70$ with students' algebra grades the following May. What does this show?

- A. Predictive (criterion-related) validity evidence
- B. Content validity evidence
- C. Test-retest reliability
- D. Proof the test is valid for any purpose.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8.

In a statewide study funded by a private foundation, a researcher records each student's class rank (1st, 2nd, 3rd, and so on). What is the level of measurement for class rank?

- A. Ratio, since any number can be averaged.
- B. Ordinal
- C. Interval
- D. Nominal

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9. [TRANSFER — reversed stem / different variable]

Which variable is measured at the ratio level?

- A. Student ID number
- B. Temperature in °F
- C. Class rank
- D. Number of books a student read this year

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

A researcher at a college with a strong education program wants to generalize to all 4,000 teachers in a district. She emails a survey and analyzes the 150 teachers who reply. The response pleased the superintendent. What is the main limitation?

- A. It is too small; at least half the population must always be surveyed.
- B. Random assignment was not used.
- C. No limitation, because 150 is a large number.
- D. It is a volunteer sample, so results may not represent all teachers.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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11. [TRANSFER — reversed stem / different variable]

A researcher needs findings that generalize to every teacher in a district. Which sampling plan fits best?

- A.** Ask principals to nominate their best teachers.
- B.** Survey teachers at the researcher's own school.
- C.** Invite volunteers; a big enough volunteer group always represents the population.
- D.** Randomly select teachers from the full district roster.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A qualitative researcher spent a year observing a rural school that has 90 students. She wants to strengthen the credibility of her conclusions about student belonging. What is the best next step?

- A.** Remove quotes that do not fit her themes.
- B.** Triangulate interviews, observations, and documents, and have participants check her interpretations.
- C.** Compute a reliability coefficient for the interview answers.
- D.** Enlarge the study to 1,000 students to reach statistical significance.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A principal at a K-8 school ~~that just opened a new cafeteria~~ wants to understand how first-year teachers experience their mentoring. She plans in-depth interviews and will code the transcripts for themes. What kind of research is this?

Target word: research approach **Decides:** Goal is understanding experience through interviews.

A	Adjacent category	No manipulation or comparison groups.
B	Confusion set	Tallying themes does not change the interpretive design.
C	Correct	Explores meaning with open-ended, word-based data.
D	Absolute language	Mixed methods integrates quantitative and qualitative strands.

2. Answer A Concept 1 • transfer

A researcher wants to test whether weekly homework time predicts end-of-year math scores across 600 students. Which approach fits?

Target word: pick the approach from the goal **Decides:** Testing a numeric relationship across many cases.

A	Correct	Measures two variables and tests their relationship.
B	Confusion set	Studies culture through immersion.
C	Inverted operation	Explores depth, not a numeric prediction.
D	Adjacent category	Local improvement, not a test across 600 students.

3. Answer A Concept 2

A district research office ~~with four analysts~~ surveys 1,200 parents about homework load. Several results are surprising, so the office then interviews 15 parents to explain them. Which design is this?

Target word: mixed-methods design **Decides:** Quantitative first, qualitative second to explain.

A	Correct	The qualitative phase explains quantitative results.
B	Inverted operation	Exploratory starts with qualitative data.
C	Confusion set	Convergent collects both strands at the same time.
D	Dropped element	Drops the interview phase.

4. Answer C Concept 3

A third-grade teacher ~~who coaches the school's chess club~~ notices many students skip their reading logs. She tries a choice-board routine for six weeks, tracks log completion, reflects on the results, and adjusts the routine. What best describes her study?

Target word: type and reach of the study **Decides:** A practitioner cycling through plan-act-observe-reflect in her own class.

A	Absolute language	One class with no control group cannot prove a general effect.
B	Confusion set	No survey; she tracks behavior across cycles.
C	Correct	Action research serves the setting where it happens.
D	Adjacent category	Studying one's own setting is the defining feature of action research.

5. Answer B Concept 4

In a district that serves 9,000 students, two teachers score every essay with the same writing rubric. Their scores often differ by two points or more. Which kind of reliability is weak?

Target word: reliability type **Decides:** Two scorers disagree.

A	Adjacent category	That is agreement among items on one test.
B	Correct	Agreement between scorers is low.
C	Confusion set	That is stability across two occasions.
D	Absolute language	A rubric helps but does not guarantee agreement.

6. Answer C Concept 4 • transfer

A researcher gives the same anxiety survey to the same students two weeks apart and correlates the two sets of scores. What is being checked?

Target word: name the property from the design **Decides:** Same measure, same people, two times.

A	Confusion set	Needs two scorers, not two occasions.
B	Dropped element	Drops the two-occasion design; it uses one sitting.
C	Correct	Stability across time.
D	Adjacent category	Validity asks what scores mean, not whether they repeat.

7. Answer A Concept 5

An algebra-readiness test from a vendor that also publishes textbooks is given in August. Scores correlate at $r = .70$ with students' algebra grades the following May. What does this show?

Target word: type of validity evidence **Decides:** Scores relate to a later outcome.

A	Correct	Test scores forecast a later criterion.
B	Adjacent category	Content validity comes from reviewing item coverage.
C	Confusion set	Grades are a different measure, not a retest.
D	Absolute language	Validity is for a specific use and builds from several kinds of evidence.

8. Answer B Concept 6

In a statewide study funded by a private foundation, a researcher records each student's class rank (1st, 2nd, 3rd, and so on). What is the level of measurement for class rank?

Target word: measurement level **Decides:** Ordered, but gaps between ranks are not equal.

A	Absolute language	Ratio needs equal gaps and a true zero.
B	Correct	Order is meaningful; distances are not.
C	Adjacent category	Assumes equal gaps between ranks.
D	Confusion set	Ranks have order; nominal categories do not.

9. Answer D Concept 6 • transfer

Which variable is measured at the ratio level?

Target word: find the ratio variable **Decides:** True zero plus equal intervals.

A	Dropped element	Numbers used as labels are nominal.
B	Adjacent category	Interval: 0 °F is not an absence of heat.
C	Confusion set	Ordinal.
D	Correct	Zero means none, and 10 is twice 5.

10. Answer D Concept 7

A researcher ~~at a college with a strong education program~~ wants to generalize to all 4,000 teachers in a district. She emails a survey and analyzes the 150 teachers who reply. ~~The response pleased the superintendent.~~ What is the main limitation?

Target word: limit on generalizing **Decides:** Self-selected volunteers, not a random sample.

A	Absolute language	Random samples far smaller than half can generalize well.
B	Confusion set	Assignment matters for experiments; sampling decides generalization.
C	Dropped element	Drops how the sample was chosen.
D	Correct	Non-probability samples limit generalization.

11. Answer D Concept 7 • transfer

A researcher needs findings that generalize to every teacher in a district. Which sampling plan fits best?

Target word: pick the sampling plan from the goal **Decides:** Every teacher needs a known chance of selection.

A	Confusion set	Purposive selection, useful for depth, not generalization.
B	Adjacent category	Convenience sample of one site.
C	Absolute language	Size cannot fix self-selection bias.
D	Correct	A probability sample supports generalization.

12. Answer B Concept 8

A qualitative researcher spent a year observing a rural school ~~that has 90 students~~. She wants to strengthen the credibility of her conclusions about student belonging. What is the best next step?

Target word: credibility in qualitative work **Decides:** Qualitative quality comes from triangulation and member checks.

A	Inverted operation	Discarding disconfirming evidence weakens credibility.
B	Correct	Triangulation and member checking build credibility.
C	Confusion set	A psychometric tool, not a credibility strategy for interpretation.
D	Adjacent category	Applies a quantitative standard to qualitative work.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	3	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	12	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Quantitative vs. qualitative research	Quantitative research tests relationships or effects with numeric data; qualitative research explores meaning and experience with words, observation, and documents.	Labeling a study by its topic or by whether anything gets counted.	“Understand how ... experience” vs. “test whether ... predicts.”
2	Mixed-methods designs	Explanatory sequential = quantitative first, then qualitative to explain; exploratory sequential = qualitative first; convergent = both at once, then compare.	Calling any study with two data sources mixed methods, or reversing the sequence names.	The order in which surveys and interviews happen and why.
3	Action research	Practitioners study a problem in their own setting through cycles of plan, act, observe, and reflect; findings improve local practice and are not meant for broad generalization.	Treating action research as proof that an approach works everywhere.	A teacher trying and adjusting a change in her own classroom.
4	Types of reliability	Test-retest = stable over time; internal consistency = items agree; inter-rater = scorers agree.	Naming the wrong type for the design described.	Two occasions, many items, or two scorers.
5	Types of validity evidence	Content = items cover the domain; criterion (predictive/concurrent) = scores relate to an outcome measure; construct = scores behave as the theory expects.	Treating one piece of validity evidence as proof for every use.	A correlation with a later outcome, or an expert review of items.
6	Levels of measurement	Nominal = categories; ordinal = ordered ranks; interval = equal gaps, no true zero; ratio = equal gaps and a true zero.	Treating ranks as equal-interval data or any number as ratio.	Ranks, °F, counts, or ID codes.
7	Sampling and generalization	Only probability (random) samples support statistical generalization to a population; convenience and volunteer samples limit it.	Believing a large volunteer sample is representative.	“Generalize to all ...” with how participants were chosen.
8	Qualitative trustworthiness	Qualitative credibility comes from triangulation, member checking, prolonged engagement, and an audit trail, not statistical significance.	Applying quantitative quality standards to qualitative work.	“Strengthen the credibility of qualitative findings.”

Course scope (public description): The Research Foundations course focuses on the essential concepts in educational research, including quantitative, qualitative, mixed, and action research. This course also teaches students concepts about measurement and assessment, as well as strategies for obtaining warranted research results.